

CHOPOK, Slovakia

WMO: 119160

Lat: 48.9433N

Lon: 19.5892E

Elev: 2007

StdP: 79.43

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.6	-16.9	-31.6	0.3	-8.7	-27.5	0.4	-7.2	27.9	-9.2	25.5	-8.2	10.2	330	1.060

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	16.1	11.9	14.7	11.3	13.5	10.7	13.1	15.0	12.1	13.8	11.3	12.9	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	11.5	13.7	11.4	10.8	12.9	10.5	10.1	12.1	43.6	15.1	40.7	13.8	38.2	13.1	15.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.9	18.7	DB	-21.7	18.4	2.6	1.5	-23.6	19.4	-25.1	20.3	-26.6	21.1	-28.5	22.2	(4)
(5)			WB	-22.2	14.4	2.6	0.8	-24.1	15.0	-25.6	15.5	-27.0	15.9	-28.9	16.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-8.1	-8.2	-6.3	-1.5	3.2	6.7	8.5	8.6	4.4	0.8	-3.1	-6.7	(6)
(7)		DBStd	7.44	4.56	4.47	4.47	4.42	3.96	3.94	3.51	3.39	3.70	4.76	4.71	4.71	(7)
(8)		HDD10.0	3747	560	508	504	345	213	112	72	66	169	287	393	518	(8)
(9)		HDD18.3	6723	818	742	763	595	469	349	306	301	417	544	643	776	(9)
(10)		CDD10.0	65	0	0	0	0	2	12	24	24	2	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.4	9.6	9.3	9.4	8.6	7.4	6.9	7.1	6.2	7.4	8.4	10.0	10.2	(14)
(15)	Precipitation	PrecAvg	959	57	59	60	67	99	106	127	89	81	79	75	59	(15)
(16)		PrecMax	1328	173	281	160	166	231	175	231	230	164	149	148	149	(16)
(17)		PrecMin	626	6	8	9	10	38	32	34	22	17	6	10	15	(17)
(18)		PrecStd	159	37	51	39	35	45	38	57	49	43	41	37	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	2.8	4.6	8.8	13.8	16.9	17.9	18.3	13.8	12.3	10.2	5.4	(19)
(20)			MCWB	-1.8	-1.8	0.0	4.6	9.1	12.3	13.1	13.0	9.9	4.9	3.9	-1.0	(20)
(21)		2%	DB	1.5	0.5	2.1	6.8	11.6	15.1	16.1	16.1	12.0	9.9	5.7	2.5	(21)
(22)			MCWB	-2.2	-2.3	-0.6	3.6	8.0	11.3	12.1	12.0	8.7	4.7	1.9	-2.2	(22)
(23)		5%	DB	-0.3	-0.8	0.7	5.2	10.0	13.5	14.5	14.4	10.6	8.2	4.0	0.7	(23)
(24)			MCWB	-2.3	-2.3	-1.0	2.7	7.3	10.7	11.6	11.4	8.0	4.8	1.9	-2.0	(24)
(25)		10%	DB	-1.8	-2.1	-0.4	3.8	8.5	12.0	13.1	13.2	9.3	6.8	2.7	-0.5	(25)
(26)			MCWB	-3.1	-3.0	-1.5	1.8	6.5	9.7	10.9	10.7	7.3	4.7	1.2	-2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.1	1.8	5.7	10.2	14.0	14.1	14.4	10.9	8.1	5.6	1.2	(27)
(28)			MCDB	1.5	0.9	2.5	7.2	12.5	15.3	16.0	15.9	12.6	8.7	7.7	2.1	(28)
(29)		2%	WB	-0.6	-1.0	0.5	4.2	8.9	12.4	13.0	13.2	9.7	7.1	3.7	-0.1	(29)
(30)			MCDB	0.1	-0.2	1.2	6.0	10.5	14.1	14.9	15.1	11.0	7.8	4.2	0.5	(30)
(31)		5%	WB	-1.8	-1.9	-0.4	3.2	7.8	11.1	12.1	12.1	8.8	6.1	2.6	-0.9	(31)
(32)			MCDB	-1.0	-1.4	0.2	4.5	9.3	12.7	13.8	13.7	9.9	7.1	3.1	-0.1	(32)
(33)		10%	WB	-3.0	-3.1	-1.4	2.3	6.7	10.1	11.2	11.1	7.8	5.2	1.5	-1.9	(33)
(34)			MCDB	-2.0	-2.4	-0.5	3.5	8.2	11.6	12.9	12.8	8.9	6.2	2.3	-1.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	4.5	4.5	4.3	4.2	4.4	4.4	4.0	4.1	3.9	4.3	(35)	
(36)			MCDBR	5.5	5.2	5.1	5.1	5.6	6.0	5.6	5.5	5.1	4.7	4.1	5.0	(36)
(37)			MCWBR	4.5	4.4	4.2	3.3	3.9	4.8	3.8	4.2	4.3	4.4	3.8	3.8	(37)
(38)		5% WB	MCDBR	5.0	4.9	4.5	4.2	4.9	5.1	5.0	4.9	4.4	4.0	3.7	4.7	(38)
(39)			MCWBR	4.9	4.8	4.1	3.0	3.9	4.7	3.9	4.2	4.0	4.2	3.8	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.231	0.249	0.280	0.320	0.319	0.322	0.330	0.327	0.306	0.284	0.264	0.232	(40)	
(41)		taud	2.404	2.380	2.398	2.416	2.447	2.453	2.418	2.439	2.493	2.548	2.542	2.460	(41)	
(42)		Ebn at Noon	867	924	942	931	941	937	923	912	903	873	821	822	(42)	
(43)		Edh at Noon	62	82	96	106	107	107	110	103	87	70	56	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.69	2.84	4.17	4.99	5.39	5.20	4.71	3.34	2.04	1.04	0.73	(44)	
(45)		RadStd	0.15	0.27	0.47	0.47	0.62	0.50	0.55	0.45	0.48	0.27	0.14	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		+0.56	N/A	N/A	+0.92	+0.51	+0.44	-125	-176	+88	+39

Nomenclature: See separate page